A Physical Feasibility Study of a Spider-Man-Like Grapple-Line Swinging Suit: Pendular Urban Locomotion with Tethered Anchors

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Abstract

A human can, in principle, cross a city by swinging on launched tethers, but only within tight limits set by tether tension, anchor strength and human g-tolerance. We model an 85 kg user on a 30 m tether as a variable-length pendulum with quadratic air drag, then follow each release with a drag-affected ballistic flight. Starting from a horizontal tether, the user reaches 21.9 m/s at the bottom of the arc and the tether carries 2.6 times body weight (3.0 without drag). The best release point is about 40° past the bottom, which carries the user roughly 82 m horizontally per swing with drag (102 m in vacuum). Reeling in 4 m of line near the bottom of the arc, the mechanism by which a child pumps a playground swing, adds about 10 kJ per swing and lifts release speed by 25%, but demands roughly 20 kW of peak winch power. The binding constraints are not the swing itself: they are anchors able to hold 8–10 kN on arbitrary building surfaces, winch power density, and catch loads if the tether goes slack. We size each subsystem, compare five anchor technologies, and list the failure modes that would have to be engineered out before any human trial. The concept is inspired by the web-swinging of the fictional character Spider-Man; this paper treats only the real physics.

Keywords: pendulum dynamics; variable-length pendulum; tether; grappling hook; swing pumping; aerodynamic drag; fall arrest; anchor mechanics

1 Introduction and scope

Pendular locomotion is old: gibbons brachiate, trapeze artists swing, and rope-access workers and climbers rely daily on the same tension and energy relations used here. Fiction has popularized a far more ambitious version: the comic-book and film character Spider-Man shoots a sticky line at a skyscraper and swings across a city at highway speed. This paper asks which parts of that picture survive contact with real mechanics and real materials.

We treat a hypothetical wearable system, called here the *grapple-line suit*, made of a line launcher, a high-strength tether, an attaching anchor, a powered winch and a load-spreading harness. We do not propose building or testing it on people. The goal is a quantitative feasibility map: what forces arise, which subsystems are near existing technology, and which are far from it.

Scope and assumptions:

- Body mass $m = 85$ kg including suit; tether length $L = 30$ m unless stated; air density $\rho = 1.2$ kg/m^3; $g = 9.81$ m/s^2.
- Drag area $C_dA = 0.5$ m^2 for a person moving through a swing, between a skydiver's belly-down value near 0.7 m^2 and a head-down value near 0.2 m^2 (Hoerner 1965).

- Anchors are fixed points on building faces or rooftop edges. The tether is massless and inextensible in Sections 3, 5 and 6, and elastic in Section 4. Motion is planar.
- All simulations integrate the equations of motion numerically (adaptive Runge–Kutta, step at most 2 ms, relative tolerance 10^{-8}). The script is deposited with this paper.

2 System architecture

The suit has five subsystems in series; the weakest sets the safe load for the whole chain. Each must carry the design load derived in Sections 3 and 4: about 2.5–2.8 kN steady peak, and 8–10 kN with a safety factor of 3–4, the margin used in rope-access practice.

Table 1: Subsystems of the grapple-line suit and their requirements. Masses and strengths for the tether are approximate.

Subsystem	Function	Nearest existing technology	Key requirement
Launcher	Fires anchor plus trailing line 20–60 m	Pneumatic rescue line throwers; grapnel launchers	Range with line drag; aim; recoil < 150 N $\cdot$ s
Tether	Carries swing tension	UHMWPE braid, 4–6 mm	≥ 15 kN break strength at ~10–25 g/m
Anchor	Holds to building	Grapnel, suction, electro-permanent magnet, powder-actuated pin	$\geq$ 8–10 kN; sets in $<$ 0.5 s; releasable
Winch	Reels line, injects energy, retracts	Powered rope ascenders	2–3 kN at 2–10 m/s (5–25 kW)
Harness	Spreads load into pelvis and thighs	Full-body fall-arrest and parachute harnesses	≤ 3 g sustained; ≤ 6 kN arrest

Two launchers, one per arm, allow the next anchor to be set while the current tether is still loaded. That overlap is what makes continuous swinging possible rather than a series of separate drops (Fig. 1).

3 Rigid-tether pendulum dynamics

Tether tension, not speed, is the governing quantity: released from a horizontal tether, the user reaches the bottom of the arc pulling exactly $3g$, independent of tether length. Let θ be the tether angle from the vertical, positive in the direction of travel, with the anchor at the origin (Fig. 2). For a tether of constant length L with quadratic drag opposing the velocity, the equation of motion is (Taylor 2005)

$$mL\ddot{\theta} = -mg\sin\theta - \tfrac{1}{2}\rho C_d A\, L^2\, \dot{\theta}\,|\dot{\theta}|. \tag{1}$$

The radial balance gives the tension that tether, anchor and harness must carry:

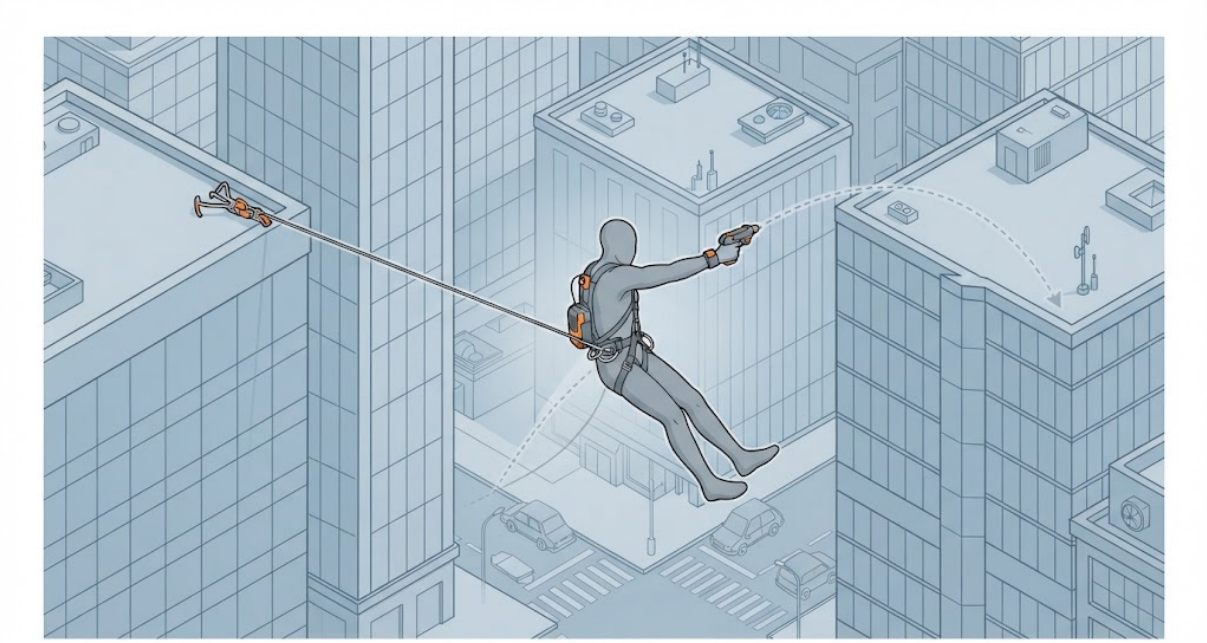


Figure 1: Conceptual illustration of the grapple-line suit at the start of a swing: the tether runs from a grapnel on one roof edge to the front of the harness, while a wrist launcher fires the next line toward the adjacent roof. Illustrative only; not to scale.

$$T = mg\cos\theta + mL\dot{\theta}^2. \tag{2}$$

Without drag, energy conservation from rest at angle θ_0 gives the speed and tension at any angle:

$$v^2 = 2gL(\cos\theta - \cos\theta_0), \qquad T = mg\,(3\cos\theta - 2\cos\theta_0). \tag{3}$$

At the bottom ($\theta = 0$) tension peaks at $T = mg(3 - 2\cos\theta_0)$. Two consequences matter for design. First, the peak load fraction depends only on the start angle, so a longer tether gives more speed ($v \propto \sqrt{L}$) without more force. Second, any start with the user already moving, such as entering a swing from a previous flight, raises the load: an entry speed v_0 adds mv_0^2/L to the bottom tension.

Table 2: Simulated bottom-of-arc speed and peak tension for $m = 85$ kg, $L = 30$ m, $C_dA = 0.5$ m^2, start from rest.

Start angle θ_0	Bottom speed, no drag (m/s)	Bottom speed, drag (m/s)	Peak T/mg, no drag	Peak T/mg, drag
30°	8.9	8.6	1.27	1.25
45°	13.1	12.4	1.59	1.53
60°	17.2	16.0	2.00	1.87
75°	20.9	19.1	2.48	2.25
90°	24.3	21.9	3.00	2.64

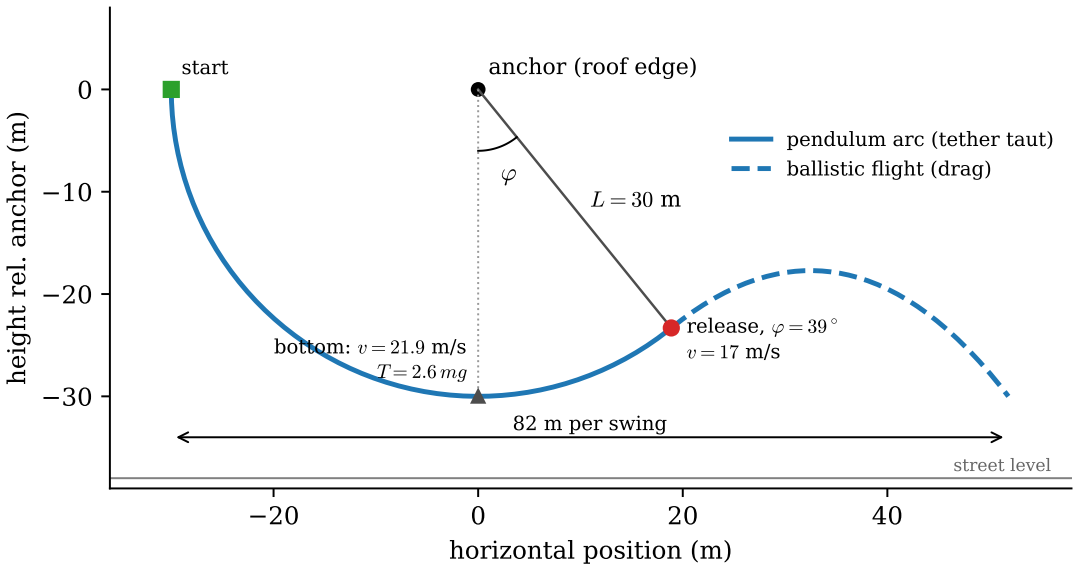


Figure 2: Geometry of one swing, computed from Eqs. (1)–(2) and (8) with drag. The user starts with the tether horizontal, passes the bottom of the arc at 21.9 m/s with tether tension 2.6 mg, releases at the optimal angle $\varphi = 39°$ and flies ballistically back to the altitude of the arc bottom, 82 m from the start.

The no-drag columns of Table 2 reproduce Eq. (3) to three figures, which checks the integrator. Drag costs about 10% of bottom speed from a horizontal start; at 22 m/s it is about 145 N, or 0.17 mg, and cannot be neglected.

4 Elastic tether and catch loads

The most dangerous load is not the swing but the catch: if the user falls with slack line before it goes taut, a stiff UHMWPE tether can deliver 13–23 kN, far beyond what anchor or body can take. Model the tether as a linear spring of axial stiffness EA (force per unit strain), so $k = EA/L$. A free fall of height h followed by stretch x gives $mg(h + x) = \frac{1}{2}kx^2$, whose solution for the peak force is the standard climbing-rope result:

$$F_{\max} = mg + \sqrt{(mg)^2 + 2\,mg\,EA\,f}, \qquad f = h/L. \tag{4}$$

The fall factor f, not the absolute fall height, sets the peak force, so the same slack ratio is equally dangerous on a 10 m or a 50 m line. A dynamic climbing rope has EA of roughly 20–25 kN; this value reproduces the 8–9 kN impact forces typical of ropes certified to the UIAA 101 / EN 892 drop test (UIAA, n.d.; European Committee for Standardization, n.d.).

Table 3: Peak catch force from Eq. (4) for $m = 85$ kg, undamped energy-balance model.

Tether type	Approx. EA (kN)	$F_{\max}$ at $f = 0.3$ (kN)	$F_{\max}$ at $f = 1.0$ (kN)
Dynamic kernmantle	20	4.1 (4.9 g)	6.7 (8.0 g)
Semi-static	60	6.4 (7.6 g)	10.9 (13.0 g)
UHMWPE braid	300	13.1 (15.7 g)	23.2 (27.8 g)

This creates a design tension. A stiff UHMWPE line is light, strong and keeps the swing geometry predictable, but it is unforgiving in a slack catch. The practical answer is a hybrid: a UHMWPE main line in series with a short energy-absorbing element at the harness (a tear-web or dynamic-rope section, as in fall-arrest lanyards), plus winch logic that never lets the line go slack. Pre-tensioning the line before stepping off a ledge, so that f stays near zero, removes the catch load entirely.

5 Energy injection by reeling, and the power budget

Without an energy source, every swing ends lower than it began, because drag removes energy and each new anchor must sit above the user. Reeling line in near the bottom of the arc restores it: shortening the tether from 30 m to 26 m over the $\pm 15^\circ$ around the bottom adds about 10 kJ and raises release speed by 25%. This is the swing-pumping mechanism analysed by Wirkus, Rand and Ruina (Wirkus, Rand, and Ruina 1998), here driven by a motor rather than by legs.

With a variable length $L(t)$, angular momentum about the anchor obeys

$$\frac{d}{dt}\left(mL^2\dot{\theta}\right) = -mgL\sin\theta + L\,D_t, \tag{5}$$

where D_t is the tangential drag force; expanding the left side yields the Coriolis-like term $2mL\dot{L}\dot{\theta}$. At the bottom of the arc gravity exerts no torque about the anchor, so a quick pull conserves angular momentum, $L_1 v_1 = L_2 v_2$, and kinetic energy grows by the factor

$$\frac{K_2}{K_1} = \left(\frac{L_1}{L_2}\right)^2, \tag{6}$$

which is 1.33 for 30 m to 26 m. The energy is supplied by the winch, which does work $W = \int T\,(-dL)$. Because the pull happens where tension is highest, it is the most effective place to add energy and also the most demanding in force.

Table 4: Effect of reeling 4 m of line linearly in θ between -15° and $+15^\circ$; start from a horizontal tether, release at 40°.

Case	Release speed (m/s)	Peak T/mg	Winch work (kJ)	Peak winch power (kW)
No reel, drag	16.8	2.64	0	0
Reel, drag	21.0	3.26	9.9	19.3
Reel, no drag	25.7	3.96	11.5	26.8

In the no-drag case the winch work equals the gain in mechanical energy to within 0.1 kJ, which checks the variable-length integration.

Power budget. Energy is easy; power is hard. At about 12 kJ of electrical energy per swing (assuming 85% drivetrain efficiency), a lithium-ion pack near 250 Wh/kg stores enough for roughly 75 swings per kilogram of cells. But the 4 m pull takes only about 0.4 s, so peak power reaches 19 kW. Delivering that from batteries alone needs on the order of 5–10 kg of high-discharge cells. Three mitigations reduce it: reel over a wider arc (power scales roughly

with the inverse of pull time), buffer bursts through a supercapacitor bank, or accept a smaller pump per swing.

6 Release and ballistic flight

Releasing at the bottom of the arc, where speed is highest, is the wrong choice: the velocity there is horizontal and the flight starts from the lowest point. Releasing past the bottom trades some speed for upward launch angle. At release angle φ past the bottom the velocity is tangent to the arc, so the launch angle above horizontal equals φ:

$$x_r = L\sin\varphi, \quad y_r = -L\cos\varphi, \quad \mathbf{v}_r = v(\varphi)\,(\cos\varphi,\ \sin\varphi). \tag{7}$$

The free flight then follows gravity and quadratic drag:

$$\ddot{\mathbf{r}} = -g\,\hat{\mathbf{y}} - \frac{\rho C_d A}{2m}\,|\dot{\mathbf{r}}|\,\dot{\mathbf{r}}. \tag{8}$$

We score each release by the horizontal distance from the start of the swing to the point where the user falls back to the altitude of the arc bottom, which is where the next tether would need to pick them up. This metric rewards both swing travel and flight travel. A projectile launched from flat ground in vacuum goes farthest at 45°, but here speed falls as φ grows, so the optimum moves lower; drag lowers it slightly further, because it penalizes high, slow trajectories.

7 Numerical simulation results

The best release angle is about 40° past the bottom of the arc, below the 45° rule for projectiles (Fig. 3). A 1° refinement puts the optimum at 41° without drag (101.8 m) and 39° with drag (81.8 m). Total time from the start of the swing to pickup altitude is about 7.0 s with drag, an average horizontal speed near 11.7 m/s (42 km/h).

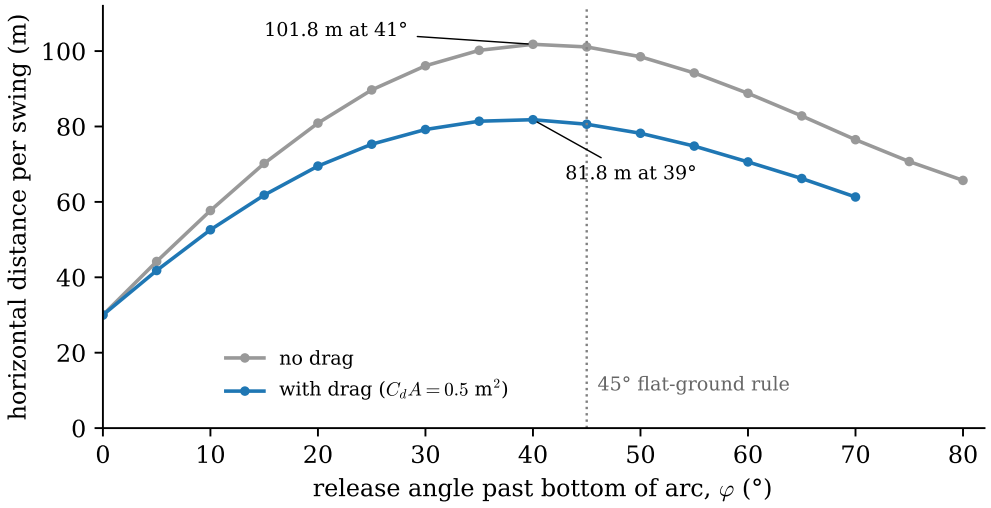


Figure 3: Horizontal distance per swing versus release angle, from Eqs. (1) and (8); $m = 85$ kg, $L = 30$ m, start from a horizontal tether. With drag the user runs out of energy near 70° and cannot reach higher release angles.

The curve is flat near its peak: with drag, releasing anywhere from 30° to 50° costs less than 5% of range. That tolerance matters, because release timing by a moving human, or by a controller reacting to anchor placement, will never be exact.

8 Anchor mechanics

The anchor is the subsystem furthest from existing technology: it must hold about 9 kN (peak swing tension of $3.3\,mg$ times a safety factor near 3.5), set in under half a second, attach to unknown surfaces, and release on command. No single technology meets all four. For a surface-pressure anchor the required contact area follows from $A = F/\sigma$, where σ is the holding stress the mechanism can sustain. For a magnetic anchor the ideal limit is the magnetic pressure (Griffiths 2017)

$$\sigma_{\max} = \frac{B^2}{2\mu_0} \approx 0.9 \text{ MPa at } B = 1.5 \text{ T}. \tag{9}$$

Table 5: Comparison of anchor technologies. Values are order-of-magnitude estimates.

Anchor type	Holding stress (approx.)	Area for 9 kN	Surfaces	Set time	Release
Grapnel over edge or rail	Set by the structure	—	Parapets, railings, beams	Instant once hooked	Easy
Electro-permanent magnet	$\leq$ 0.9 MPa; less with gaps, paint, thin plate	0.01 m^2 ideal; 0.03–0.05 m^2 real	Steel $\geq$ 10 mm	ms	Electrical pulse
Vacuum suction	$\leq$ 101 kPa; 60–80 kPa practical	0.11–0.15 m^2	Glass, polished stone, metal panels	0.1–0.5 s	Vent valve
Gecko-type dry adhesive	~100 kPa shear (animal); tens of kPa (synthetic)	0.1–0.3 m^2	Smooth, clean; shear-loaded	Contact and drag	Peel
Embedded pin or adhesive	Pins: a few kN each; adhesives: MPa after cure	Several pins, or ~0.01 m^2 bond	Concrete, masonry	Instant (pin); s–min (glue)	Destructive

Two physical points shape anchor design. First, load direction matters as much as strength. When the anchor sits on a vertical facade above the user, the swing loads it mostly in shear; dry adhesives and suction cups are several times stronger in shear than in peel (Autumn et al. 2000;

Kendall 1975), so the tether should leave the pad parallel to the wall. Second, tension cycles between about $0.3\,mg$ and $3.3\,mg$ over each arc, so a pad sees cyclic loading at roughly 0.1–0.2 Hz and must resist creep and fatigue, not just a single peak.

The fictional instant-bonding web fluid would need to reach structural-adhesive strength (of order 1 MPa) on dirty, wet or painted surfaces in under a second, then dissolve on demand. No known material does this; it is the single largest gap between the concept and reality. A realistic first system would use a grapnel or rail hook where structure allows, with a magnetic or suction head as a secondary mode on suitable surfaces.

9 Line deployment

Launching the anchor is energetically easy, about 400 J, but timing and aim are hard. Without drag, the minimum launch speed to reach a point at horizontal distance R and height h is

$$v_{\min}^2 = g\left(h + \sqrt{h^2 + R^2}\right). \tag{10}$$

For a target 40 m away at 45° elevation ($R = h = 28$ m), $v_{\min} = 26$ m/s. Trailing-line drag and spool friction roughly double the requirement, so a launch speed near 50 m/s is a reasonable design value. For a 0.3 kg anchor that is about 375 J of kinetic energy and a recoil impulse near 15 N · s, within the range of handheld pneumatic line throwers. The line is paid out from a spool rather than accelerated all at once, so its 0.5–1 kg mass adds drag and friction but not launch inertia.

Two constraints follow. Time of flight to the target is about 1.5–2 s, so the next anchor must be fired well before release from the current swing; with an arc lasting about 7 s, two alternating launchers make this possible. Aim is the harder limit: hitting a 10 cm rail at 40 m requires about 2.5 mrad of angular accuracy while the user moves at 15–20 m/s. Long, continuous targets such as roof parapets are far more forgiving than point targets, which again favours edge-hooking grapnels.

10 Human factors and safety

A trained, harnessed adult can tolerate the swing loads, which peak near $3g$ for about 1 s along the body axis (Eiband 1959). The danger lies elsewhere: in catch loads, clearance and anchor failure.

- **Harness loading.** At $3.3\,mg$ the harness carries about 2.7 kN. A sit-style or full-body harness spreads this over the thighs and pelvis; arm-held grips cannot, since sustained grip force for most adults is well under 1 kN. The tether must attach to the harness, with the arms only steering.
- **Ground clearance.** The bottom of the arc lies a full tether length below the anchor. A 30 m swing needs the anchor at least 33–35 m above the street, roughly a ten-storey building, and a clear corridor along the whole arc.
- **Wall strike.** If the anchor is not directly ahead of the user, the arc curves toward the facade. At 20 m/s, an unplanned wall contact carries about 17 kJ, a potentially fatal impact.

- **Anchor failure.** If the anchor lets go near the bottom, the user becomes a projectile at 20 m/s. Any real system needs a redundant anchor or a backup deceleration device such as a ballistic parachute, which works only above a minimum height.
- **Slack-line catches.** Section 4 shows these produce 8–28 g. Winch control must hold tension continuously and treat slack as a fault.
- **Third parties.** Launched anchors and swinging bodies endanger people below; damage to building surfaces raises legal issues in most jurisdictions.

Nothing in this paper is a basis for testing on people. Any prototype work should begin with instrumented dummies on fixed test rigs under rope-access and fall-protection standards.

11 Conclusions

Tethered swinging across a city is consistent with classical mechanics, and the swing itself is the easy part. An 85 kg user on a 30 m line reaches 22 m/s, covers about 82 m per swing at an average near 12 m/s, and experiences at most 2.6–3.3 g. The hard problems are, in order of difficulty:

1. An anchor that holds about 9 kN on arbitrary surfaces, sets in under 0.5 s and releases on command. Edge-hooking grapnels and magnetic heads come closest on suitable structure.
2. Peak winch power near 20 kW for energy injection, which pushes toward supercapacitor buffering or gentler, longer pulls.
3. Aim and timing for anchor launch at 40 m while moving at 15–20 m/s.
4. Protection against slack-line catches and anchor loss, the two failure modes with fatal consequences.

Useful next steps are a three-dimensional model with out-of-plane anchors and wall geometry, an elastic-tether swing model with damping, and bench tests of shear-loaded suction and magnetic anchors under cyclic load.

Declarations

Code availability. All numerical results are reproduced by the Python script `swing_simulation.py` (NumPy, SciPy), deposited with this paper.

AI assistance. The text, simulations and figures were drafted with the help of an AI assistant.

Competing interests. None declared.

Disclaimer. This is a theoretical study. It is not a design for use, and nothing here should be attempted by people.

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